

Portfolio | A B M Morshedul Hassan

I am a professional Urban Planner, working over a decade in designing and executing large-scale real estate and land projects. My expertise includes land use planning, urban layout design, GIS-based utility mapping by focusing on sustainability. Over the years, I've contributed to diverse projects, ranging from residential to military establishments and water network mapping.

My passion for integrating creativity with technology is reflected in my design works, GIS-based master plans, and spatial analysis of projects. I excel in tools like AutoCAD, ArcGIS, and Pix4DMapper, utilizing these to bring innovative solutions to complex urban planning issues.

Beyond my technical expertise, I'm a proud member of the Bangladesh Institute of Planners (BIP) and enlisted town planner of the City Development Authority of Dhaka (RAJUK). I am a continuous learner who wants to learn new things, collaborate and create new knowledge base.

1. During Covid-19 Pandemic i was a front line workforce in Kuwait to support planning & implementation of several field hospital and quarantine center to combat Covid-19. Picture showing one of my site while i was inspecting work progress.
2. I have a passion for designing posters and infographs. In 2014, I won the 1st prize for poster designing in a national level competition for World Town Planning Day.
3. I love hosting events. During my student life I hosted several cultural shows.



4. Our working team participating in an experience sharing event where, i was a presenter. We shared our knowledge and experience for reducing construction waste and its management.
5. In 2019, I awarded BMC Commander's Certificate of Appreciation for Best Officers.
6. Got a Certificate of Gratitude and Appreciation from the Kuwaiti Prime Minister for Confronting the Covid-19 Pandemic



Undergrad Level Thesis Projects | Jahangirnagar University

Department of Urban and Regional Planning
Jahangirnagar University, Bangladesh



Mapping the Effects of Natural Disaster on Human Displacement and Migration in Bangladesh: Special Focus on their Spatial Distribution in Dhaka

This undergraduate level thesis project was aimed at identifying effects of natural disaster on human displacement and migration in Bangladesh, to develop maps of human displacement and migration in Bangladesh and to explore the present scenario of the migrated people and their spatial distribution of settlements in Dhaka. This total thesis was my single effort.

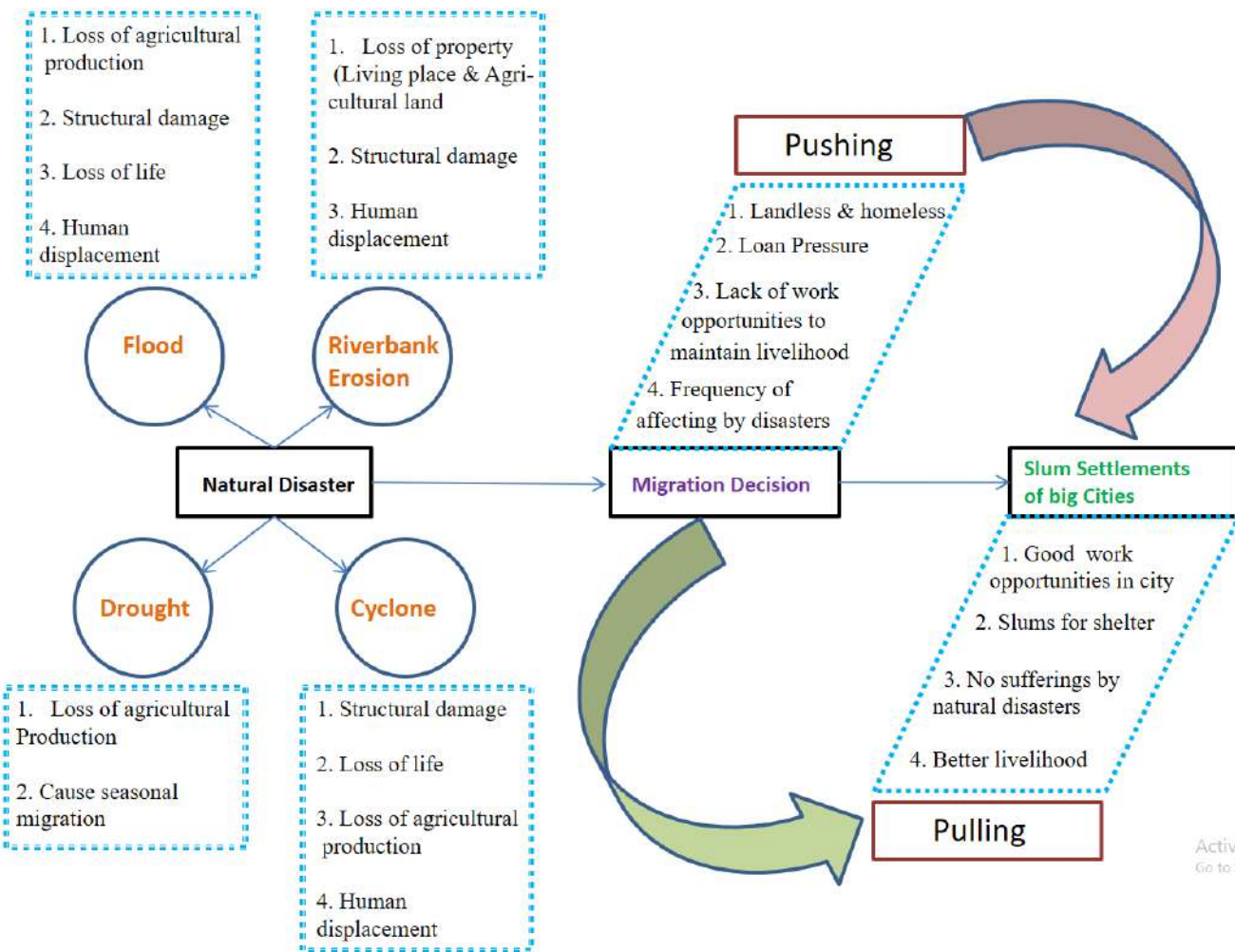


Figure illustrates impacts of natural disaster in Bangladesh

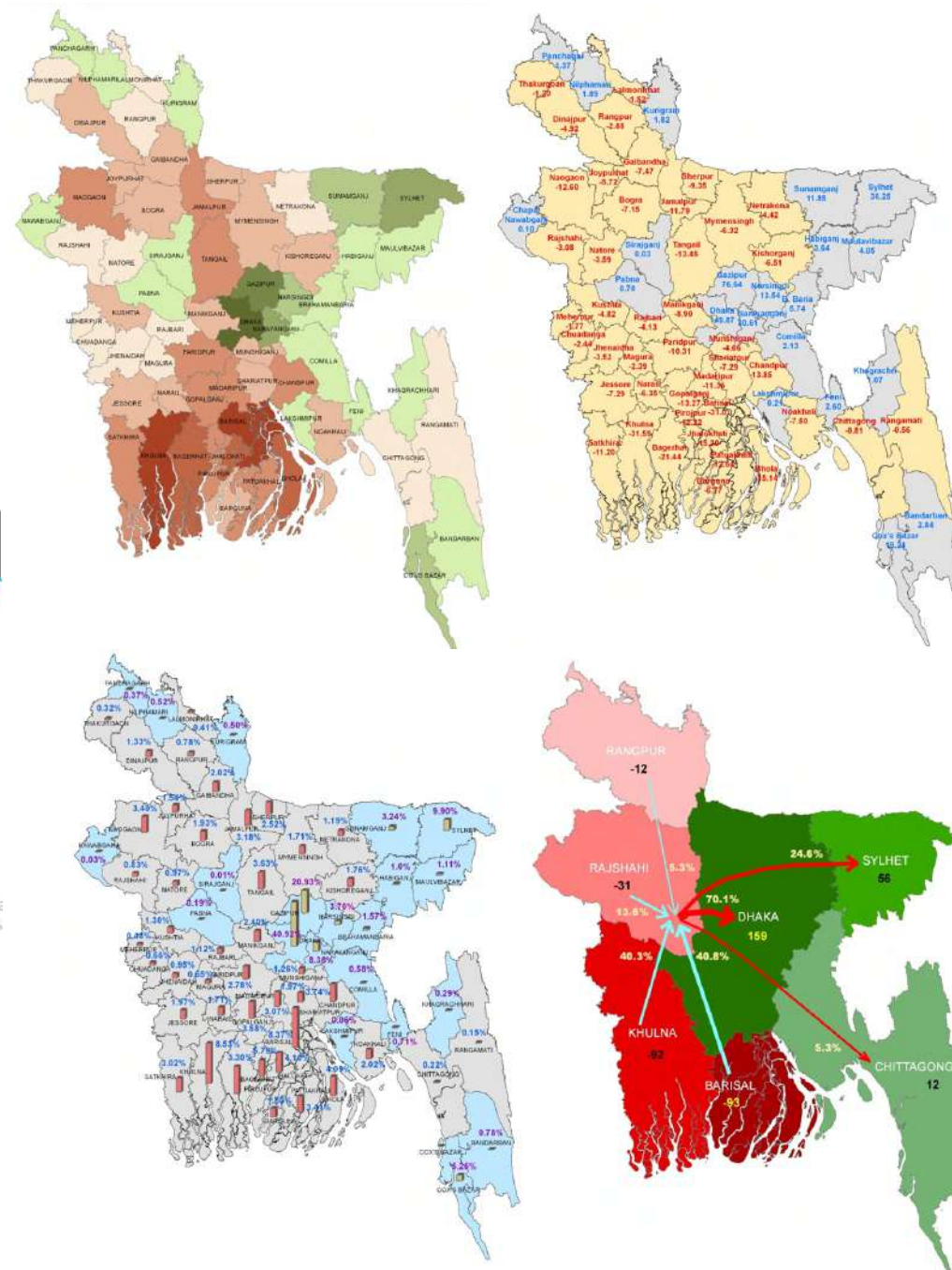
Top 5 Districts for disaster induced In Migration

District	Net migration per day	Percentage %
Dhaka	149.87	40.92%
Gazipur	76.64	20.93%
Sylhet	36.25	9.90%
Narayanganj	30.61	8.36%
Cox's Bazar	19.24	5.25%

Top 5 Districts in disaster induced Out Migration

District	Net migration per day	Percentage %
Khulna	-31.59	8.53%
Barisal	-31.0	8.37%
Bagerhat	-21.44	5.79%
Bhola	-15.14	4.09%
Chandpur	-13.85	3.74%

Given maps are showing Disaster induced internal migration in Bangladesh. District and division wise data are represented by the following maps.



In case of in migration Dhaka district is on the top of the list. Gazipur district, which is the adjacent district of Dhaka is 2nd on the list.

In case of out migration Khulna, Barisal, Bagerhat are on the top. These three districts are in coastal region of Bangladesh and mostly disaster prone area.

The following map showing the distribution of disaster migrant's settlements in Dhaka City Area. One interesting finding from this research is that, most of the migrants from Northern & Central Region districts settle themselves in northern part of Dhaka and most migrants from Southern region move to southern part of Dhaka. Three major factors have been identified for the distribution of migrants are: 1. Transportation Route, 2. Near to relatives, 3. Availability of work



Excessive migration affecting urban environment, by increasing the population density over permitted population density and creating pressure on: Urban amenities and utility services, Transportation and Urban land. Affecting the provisions of master plan by violating rules by establishing settlements in flood prone area and increased environmental pollution.



Design Procedure of an Urban Water Distribution Network & Its Performance Evaluation: A Case Study of Newly Installed Water Supply Network in Baridhara Area

This thesis was my graduate level work. This research was aimed at finding the reasons behind the malfunctioning of an age-old water distribution network in Baridhara Area, to describe the brief procedure of designing an urban water distribution network under the DWSSDP project and to evaluate the performance of a newly installed water distribution network by Dhaka WASA in Baridhara Area, Dhaka.

Reasons behind the malfunctioning of previous distribution network were: Age-old Water Distribution Network, Leakages in the System Network, Low Water Quality due to Faulty Network, Inadequate Network and Illegal House Connection, Low Capacity of Production Tube Well, Network of Small Dia, Improper Operating and Maintenance, Lack of Public Awareness.

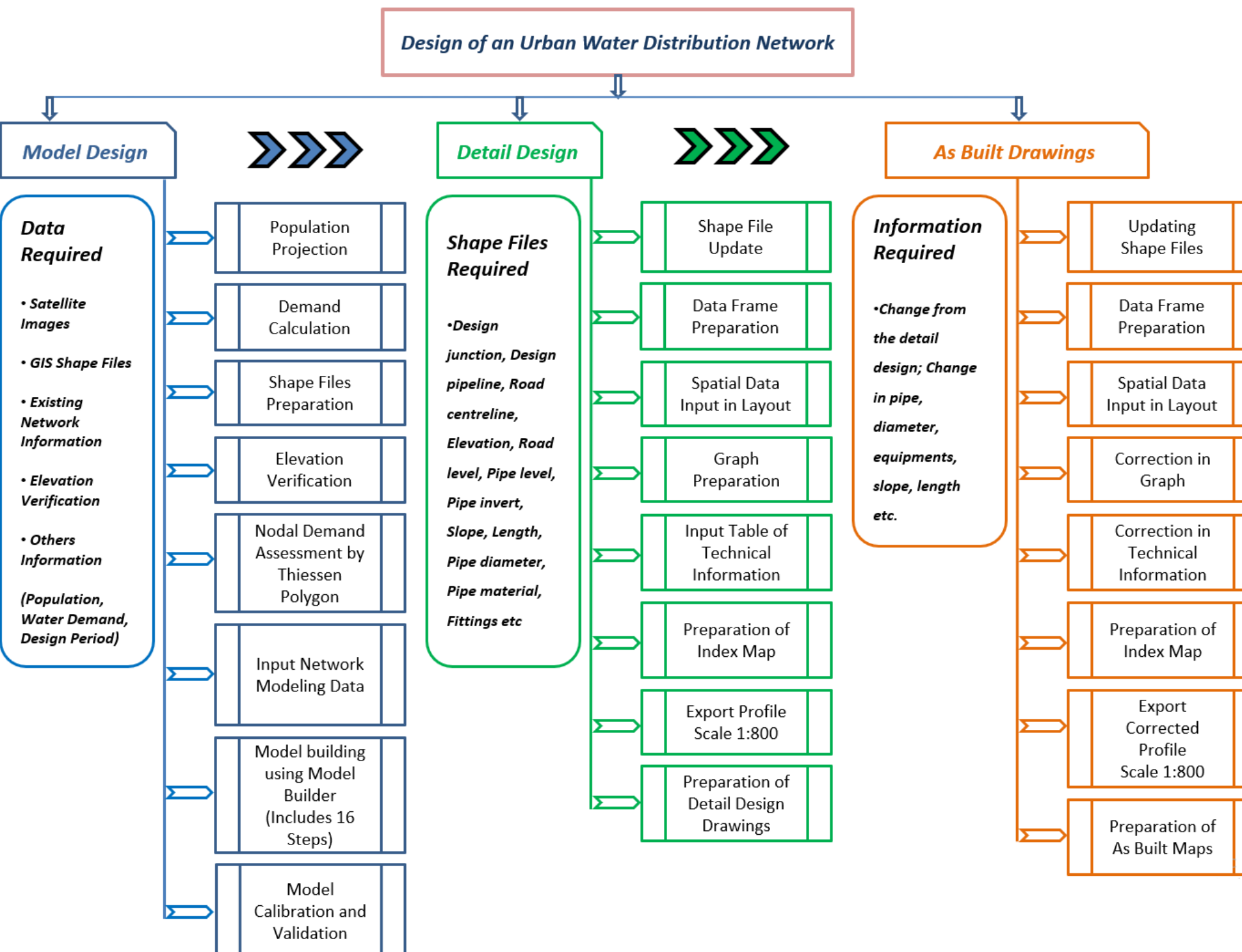
This Design Procedure is divided into three steps. They are as follows. Model Design is to create & represent digital model of real water supply condition.

This Design Procedure is divided into three steps. They are as follows. Model Design is to create & represent digital model of real water supply condition.

Detail Design made from the model design for construction purpose. As Built Drawings are the modification from the detail design after installation in real field condition.

This data driven design process made the whole project implementation more realistic and effective.

Data driven approach for designing this type of water network is very efficient. Very detail ground work makes this process easy to construct. After implementation general public given their positive opinion regarding its benefits. Considering the advantages of this project many similar projects have been implemented. In a highly dense city like Dhaka, it is beneficial for the service provider to maintain a GIS database of the network and provide better service. Public responses were good after the project implementation.



Area Name	Baridhara Residential area (Eastern portion)	
Land Area	6792 Sq. m	
Landuse Type	Residential	55.77%
	Mixed landuse	22.83%
	Commercial	21%
	Community facility	0.40%
Population	12855 Persons	
Population Growth Rate	2.12%	
Total number of House Connection	596	
Total number of Structures	744	
Area Status	High Class Residential and Mixed use Zone	
Approximate Water Consumption (Per Capita Per Day)	200 lpcd	

Private Residential Land Development Project | Master Planning and Planning Compliance

Currently i am working as a town planner with Rupayan Housing Estate Ltd and Rupayan Land Development Ltd, where i have to prepare Masterplan for townships, hotels and resorts and other plans for development. Here, i am sharing four significant plans for better understanding



Firstly, this project located in Sylhet. The master plan area is under the jurisdiction of Sylhet City Corporation Area. I have prepared this concept plan. Total project area is around 66.50 Acres. Design population is 24000. As per guidelines, required public facilities have been kept for the inhabitants. This township project will be implemented by 2030.



Rupayan city is a well known private residential land development project in Bangladesh. I am working with a team of engineers, surveyors, planners for the implementation of this project. My role is Urban Planner. My main jobs are planning, design and implementaion of the project considering planning guidelines of the city development authority of Dhaka (Rajuk). This project is implementing to provide housing for growing urban population. Total project area 213.39 acres. In this site and service scheme people got opportunity to invest in a planned neighbourhood where civic facilities with business and educational services will be provided. Total 75.27 acre land kept for Residential Purpose, Commercial land area preserved about 2.05 acres, Educational Facilities designed in 2.20 acres of land, Community facility provided in 28.01 acres, Road 28.83 acres and 1.66 acres of land kept for utilities. Design population of this project is 63000.

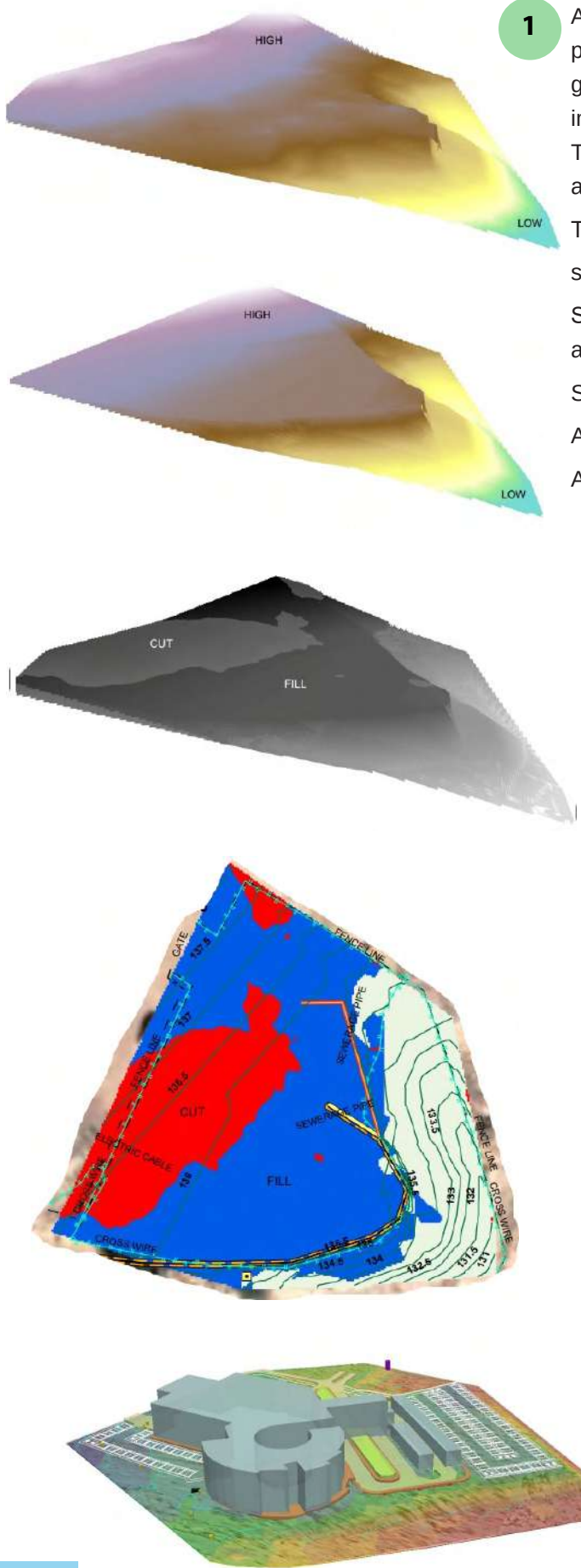


Currently, I am working on a project for tourism development in Bangladesh. It is a touristic city development project. This project will include resorts, villas, Water Park, Theme Park, Heritage park, commercial development, retail area, condominium development etc.

Considering the market survey we have allocated necessary land areas for development and prepared a land zoning map. An international Consultant firm has been hired for the detail master plan preparation for the project.

In this project there are a lot of works regarding planning compliance issues. I have to always review zoning regulation, Environmental guidelines, Local master plan rules, Tourism development guidelines etc.

Rupayan is planning for a river front township in Dhaka. This 22 acres land area is suitable for a gated community. So, I have made this plan by prioritizing water front developmnet, by providing canal connected with the river. All other community facilities have been provided as per the government guidelines. This gated community will be a healthy place for living. This township is well connected with the other areas of the city.



1 As an site planner i had to conduct physical feature survey and topographic survey for any project for the proper implementation of the project. Topographic and feature data used for analysing natural drainage of the site.

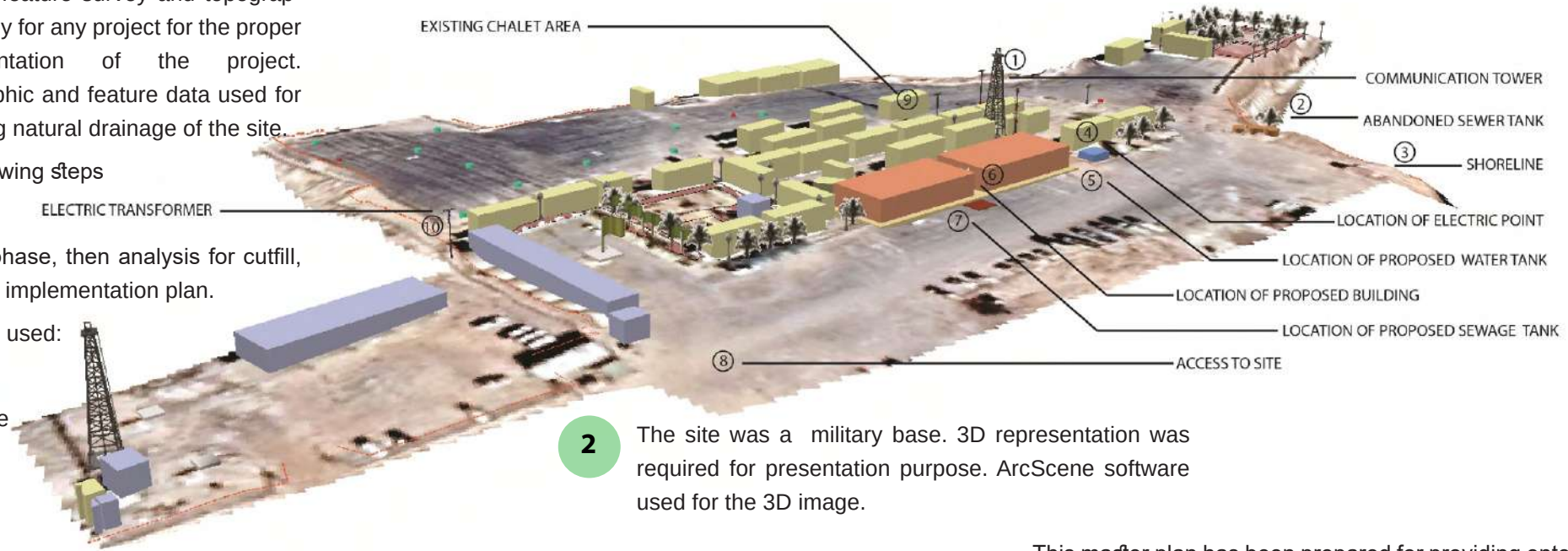
The following steps showing

Survey phase, then analysis for cutfill, and then implementation plan.

Software used:

ArcGIS,

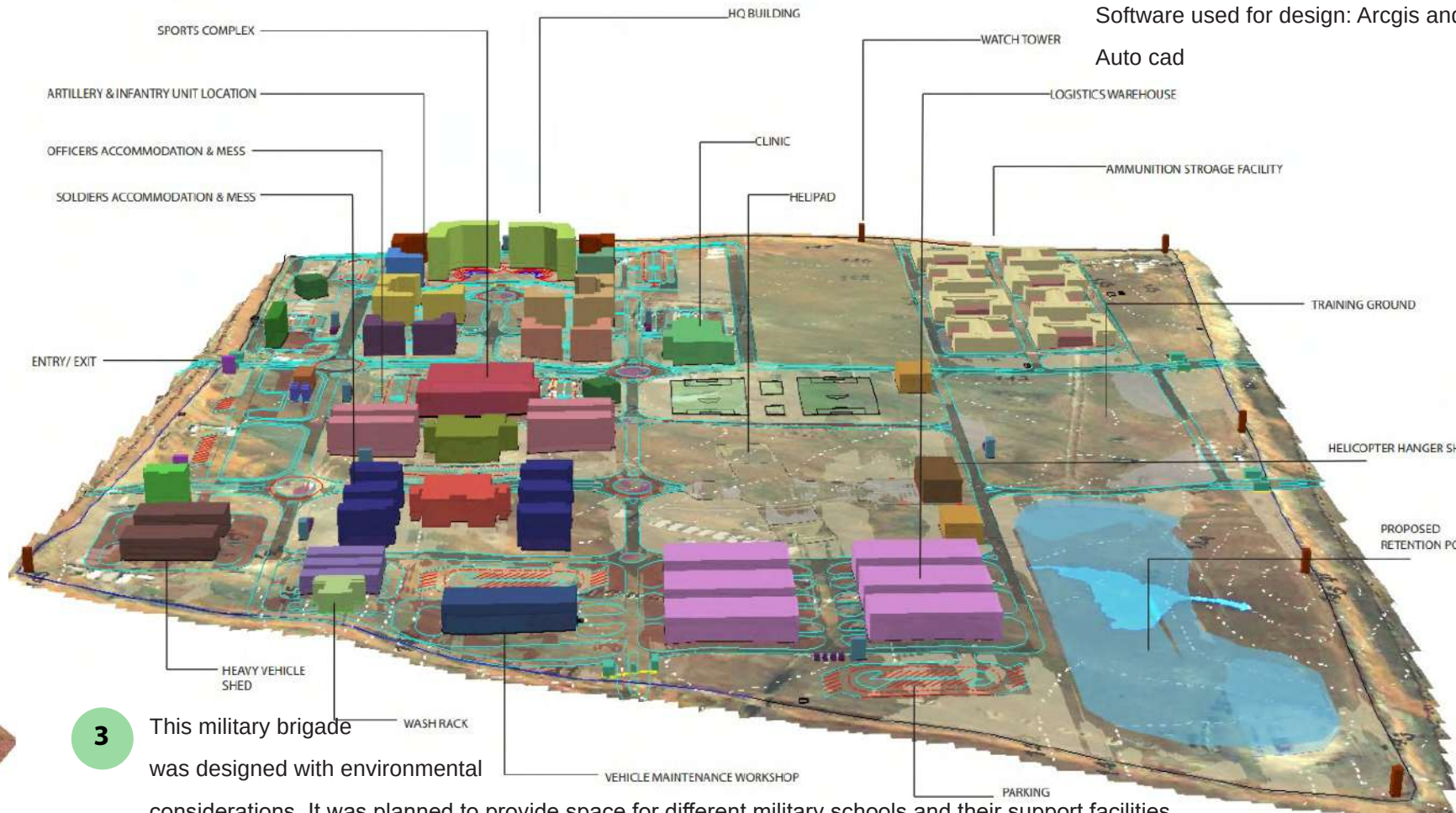
ArcScene



2 The site was a military base. 3D representation was required for presentation purpose. ArcScene software used for the 3D image.

This master plan has been prepared for providing entertainment facility for the military officers. There were proposals for club area, officers hostels, food shops, coffee shops, green areas, gardens, swimming pools etc.

Software used for design: Arcgis and Auto cad



3 This military brigade was designed with environmental considerations. It was planned to provide space for different military schools and their support facilities.



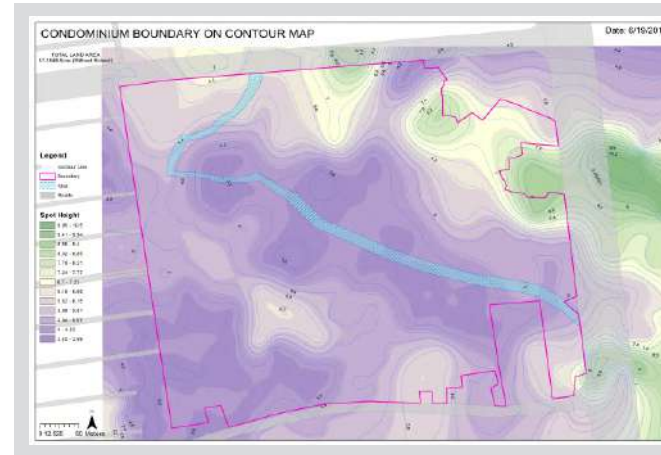
Private Residential Land Development Project | Spatial Analysis, Land use Plan, Planning Compliance

1

United City is a well known private residential land development project in Bangladesh. I worked with a team of engineers, surveyors, planners for the implementation of this project. My role was Urban Planner. My main job was planning, design and implementaion of the project considering planning guidelines of the city development authority of Dhaka (Rajuk). This project was designed to provide housing for growing urban population. Total project was designed in an area of 137.2145 acres. In this site and service scheme people got opportunity to invest in a planned neighbourhood where civic facilities with business and educational services have been provided. Total 64.20 acre land kept for Residential Purpose, Commercial land area preserved about 14.85 acres, Educational Facilities designed in 9.22 acres of land, Community facility provided in 2.83 acres, Recreation zone covered 11.717 acres, Hospital 1.50 acres, Road 31.20 acres and 1.66 acres of land kept for utilities. Design population of this project is 38,596. Several projects have been implemented for the utility (Water and Electricity) construction.

2

United condominiums is a part of United City. This residential zone is designed for 10,800 population. To support 10,800 persons basic services and amenities are designed as per standered. Total land area of the project is 61.32 acres. To give the feel of green and openness, total 25.19 acre of land has been kept for park and green. Other amenities include clinic, community center, sports club, religious center etc.



2.3

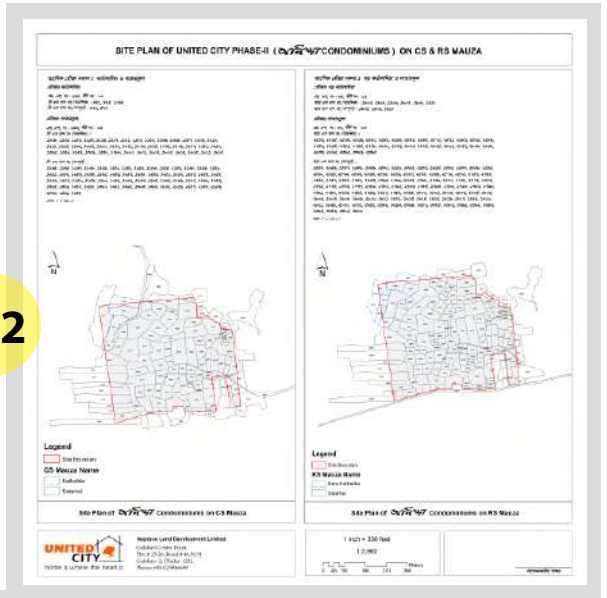
Topographic Analysis of the project area

2.1

United condominiums layout plan over satellite image.

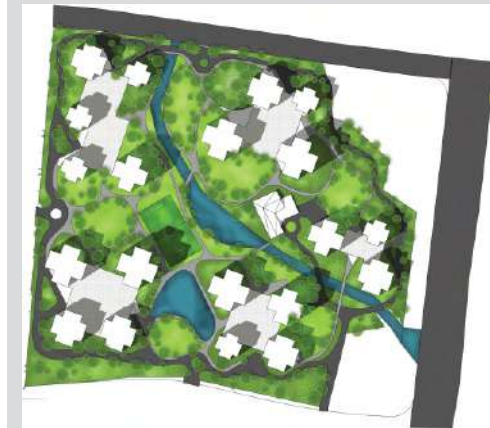
2.2

Preparation of documents for plan approval.

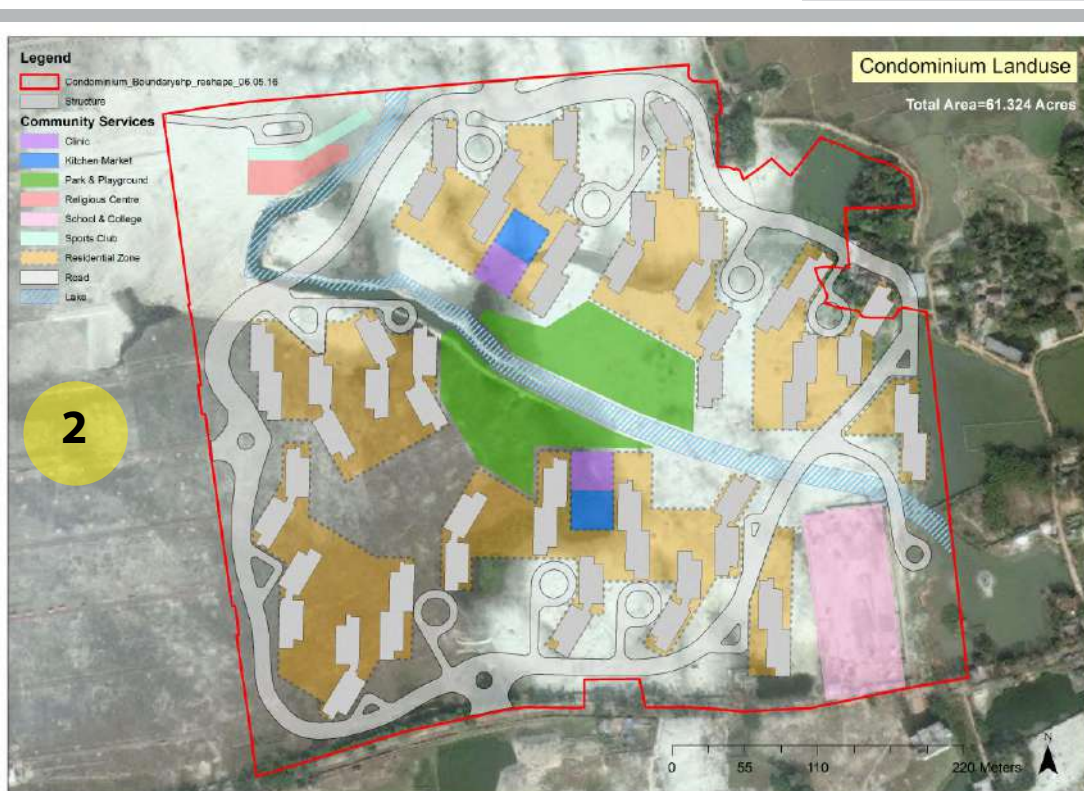


2.4

Initial Building Footprint of the Project



2



1

United City Layout Plan, Road Network Plan and Zoning areas



Dhaka Water Supply Sector Development Project | Spatial Analysis and Design Water Network



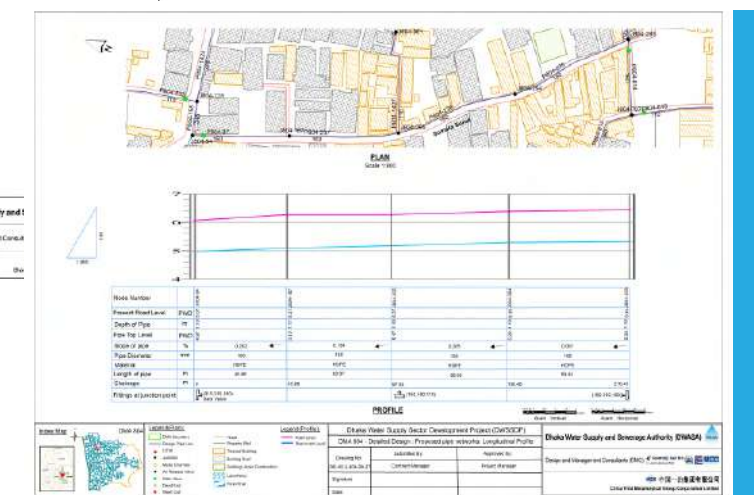
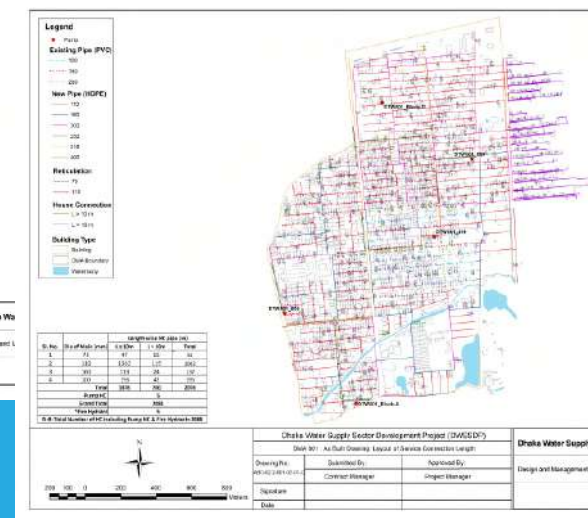
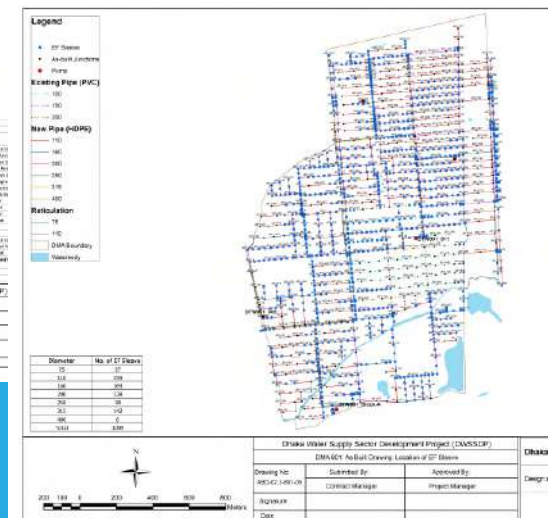
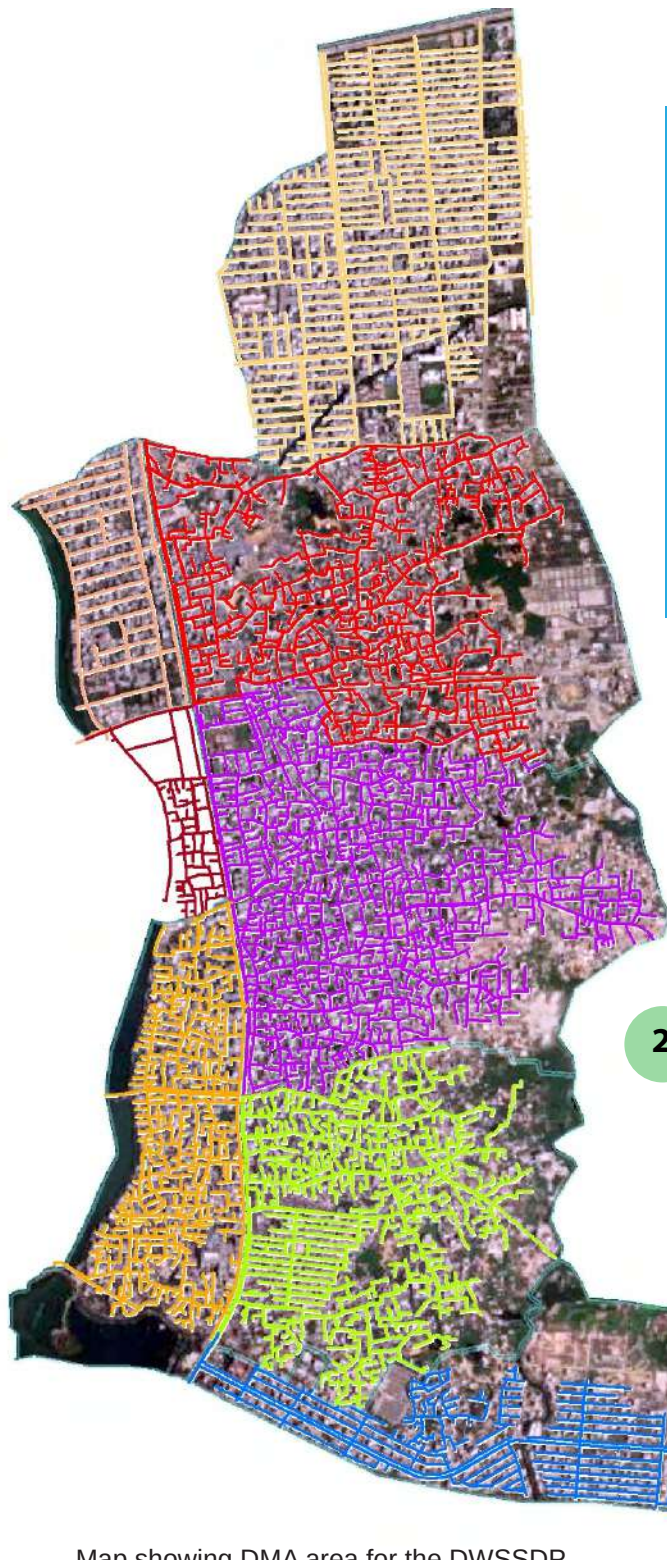
CFMCC, Bangladesh

1

The overall expected impact of the Dhaka Water Supply Sector Development Program (DWSSDP) was to contribute to sustained economic growth and public health improvement, in urban water supply sector, particularly in Dhaka Metropolitan Area, through improvement of living standard in the basic human needs by improving the water supply. I worked as an GIS professional for design and implementation of this project. A GIS team consisting ten members worked together for this project implementation.

I worked as the GIS team lead for GIS based water model database preparation, geodatabase preparation for detail design and survey data collection and update for as built preparation. Total 17 nos Detail Design Drawing, Node Sketches, Key Map, Profiles, Standard Drawings, Pipe Summary had to submit for the approval of a single district metered area (DMA). For effective calculation and data

processing, several tools based on arcpy and vbscript were developed and used. Several arctoolbox model also provided support for data processing and map preparation.



Total length of Installed Pipeline= 288 km

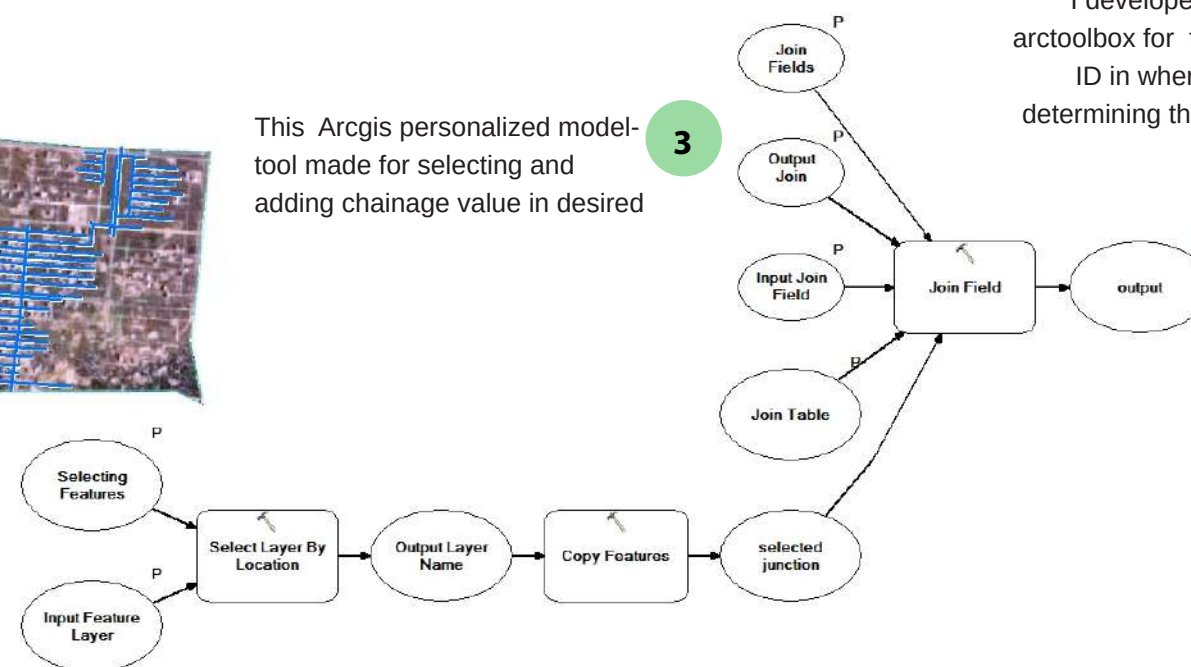
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Total No. of HC= 23,000

Construction Period= 4 Years

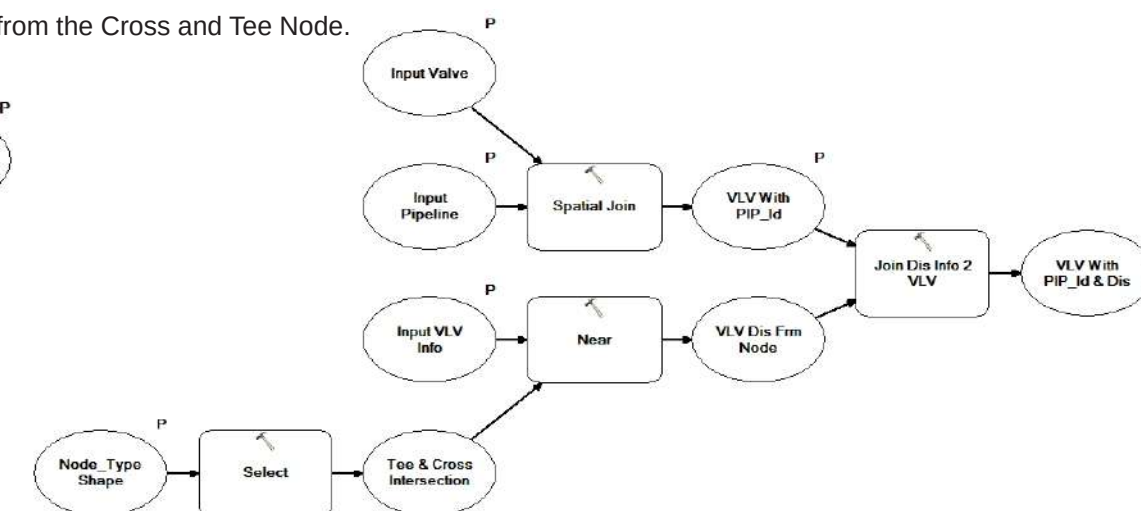
This Arcgis personalized model-tool made for selecting and adding chainage value in desired

3



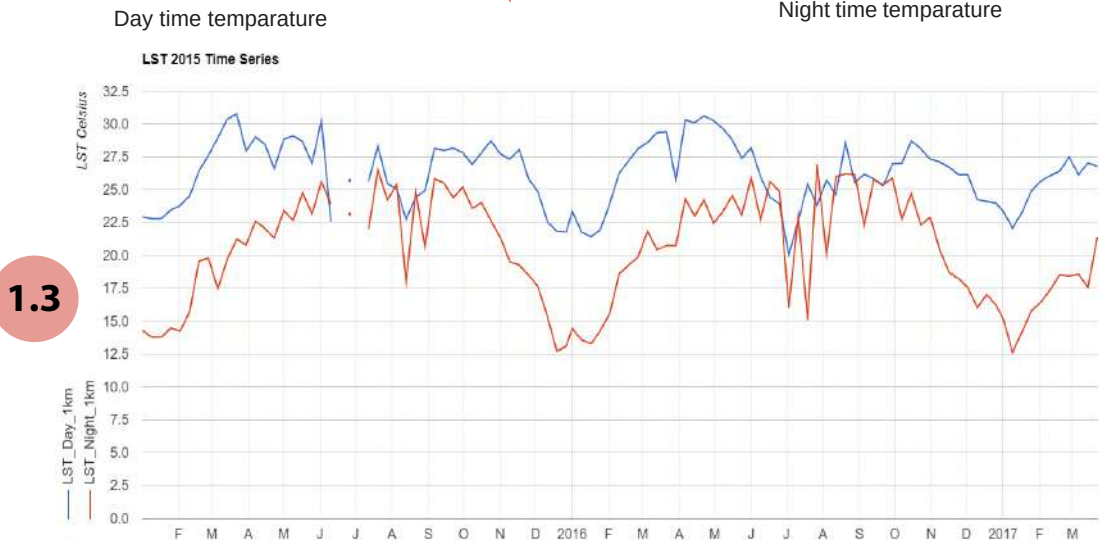
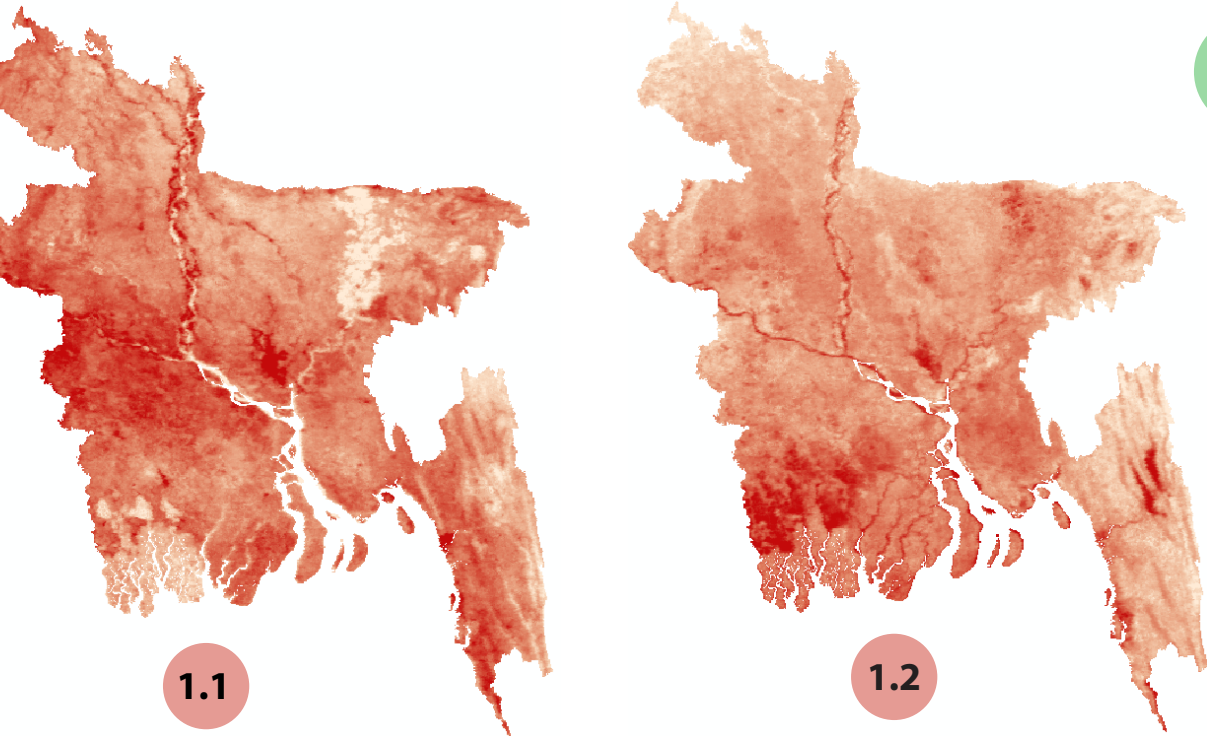
I developed this customized model using arctoolbox for for determining the actual Pipe ID in where the VLV is installed and also determining the actual distance of the Valves from the Cross and Tee Node.

4



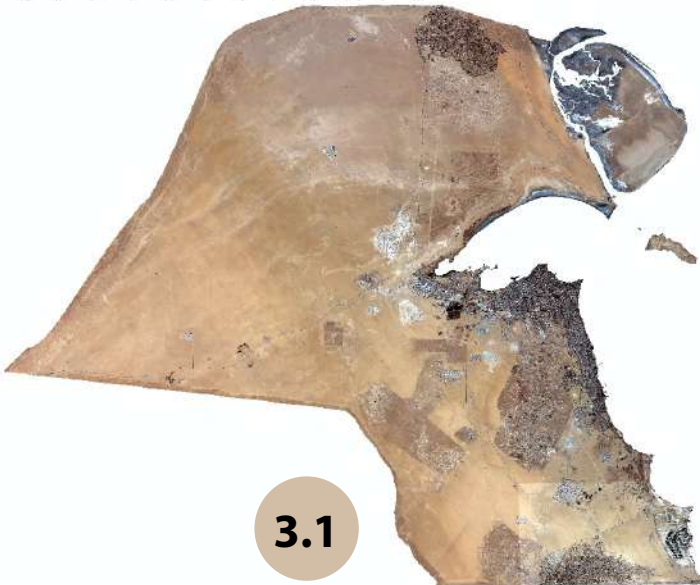
Map showing DMA area for the DWSSDP Project for water distribution network.

Remote Sensing Projects Using Google Earth Engine (GEE) | LST, NDVI & SRTM Data Analysis



1

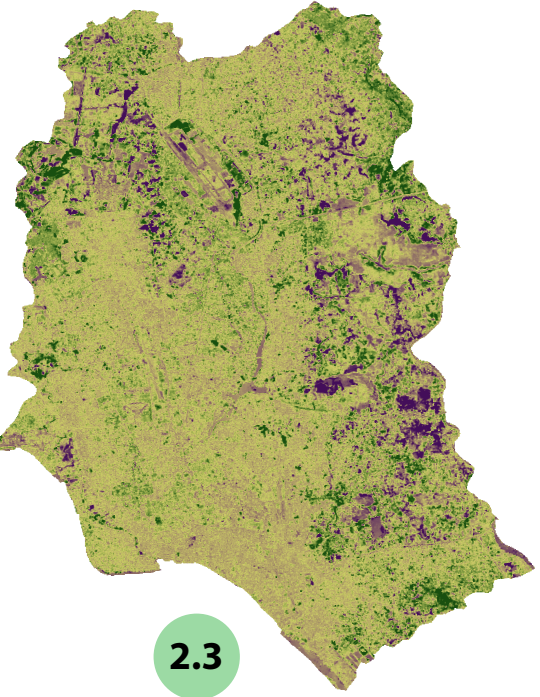
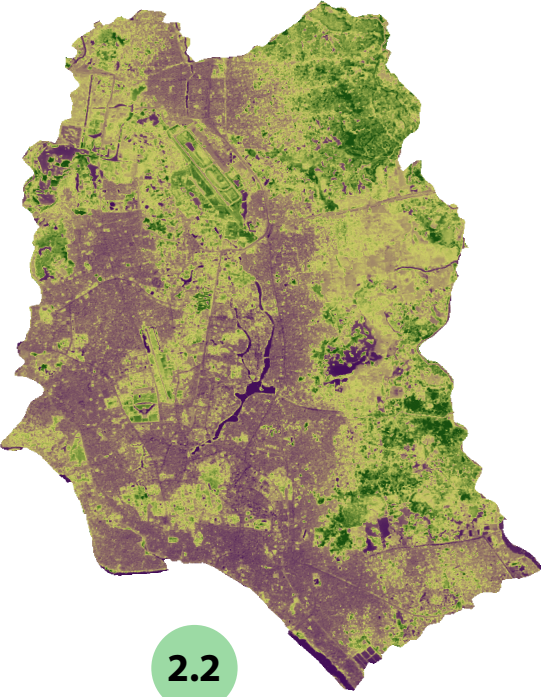
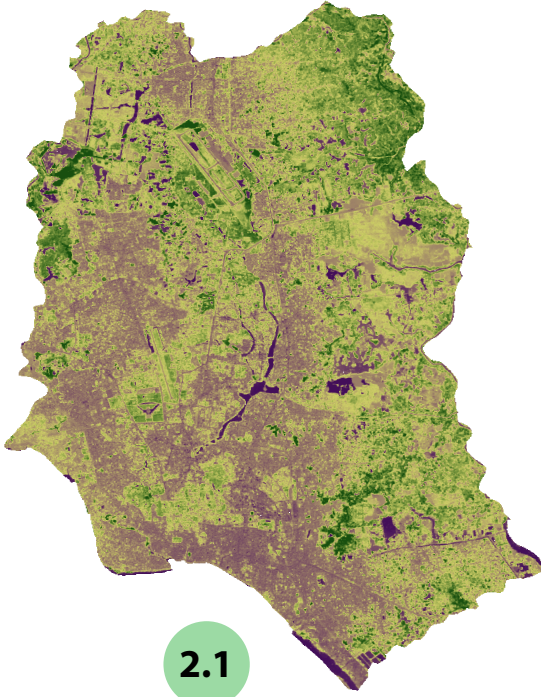
Google earth engine provides data sources for analysis land surface temperature. I have used MODIS data for analysing Land Surface temperature of Bangladesh. Image 1.1 Showing Day time temperature in degree celcius and Image 1.2 showing Night time temperature in degree celcius. And image 1.3 showing graph for the day time and night time temperature. This graph helps to find significant changes in day to night time temperature.



2

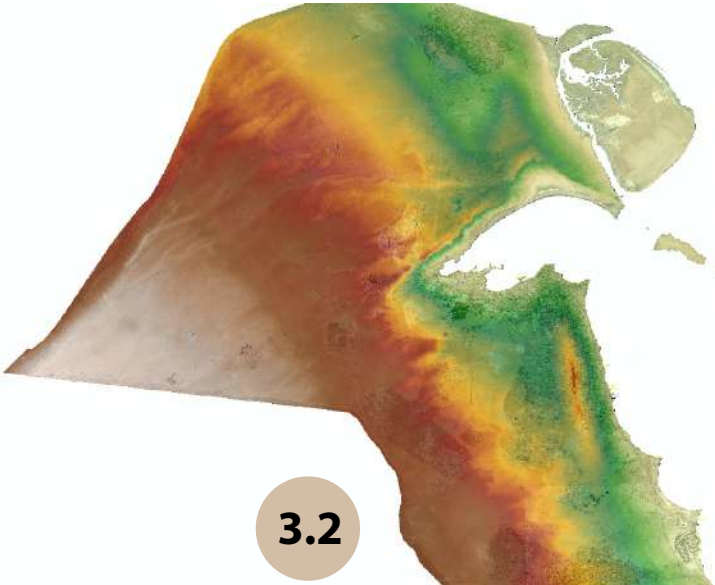
Google Earth Engine is a cloud-based geospatial analysis platform that enables users to visualize and analyze satellite images of our planet. I have used javascript codes for analysis following data for educational purpose.

The following images showing the processed Landsat8 data for the NDVI (Normalized Difference Vegetation Index) analysis in Dhaka Metropolitan Area. Image 2.1 Representing NDVI of the study area in 2015, image 2.2 illustrates NDVI in 2020 and changes over the five years period presented in image 2.3.



3

Google Earth Engine is an effective tool for remote sensing analysis. I have used google earth engine javascript code for the preparation of image number 3.1, 3.2 and 3.3. Image 3.1 representing the extraction of Landsat 8 image for Kuwait admin area. Image 3.2 showing the extraction of DEM data from Nasa SRTM database and image 3.3 representing the stream analysis of the area.



Web GIS Projects | LIS Dashboard, Utility Web Map, Geotagged Photo Web App



Web mapping is the next big thing in spatial analysis. I have learned javascript, html, css, postgis database for the implementation of web application. I have created few mapping dashboards using different javascript based plugins.



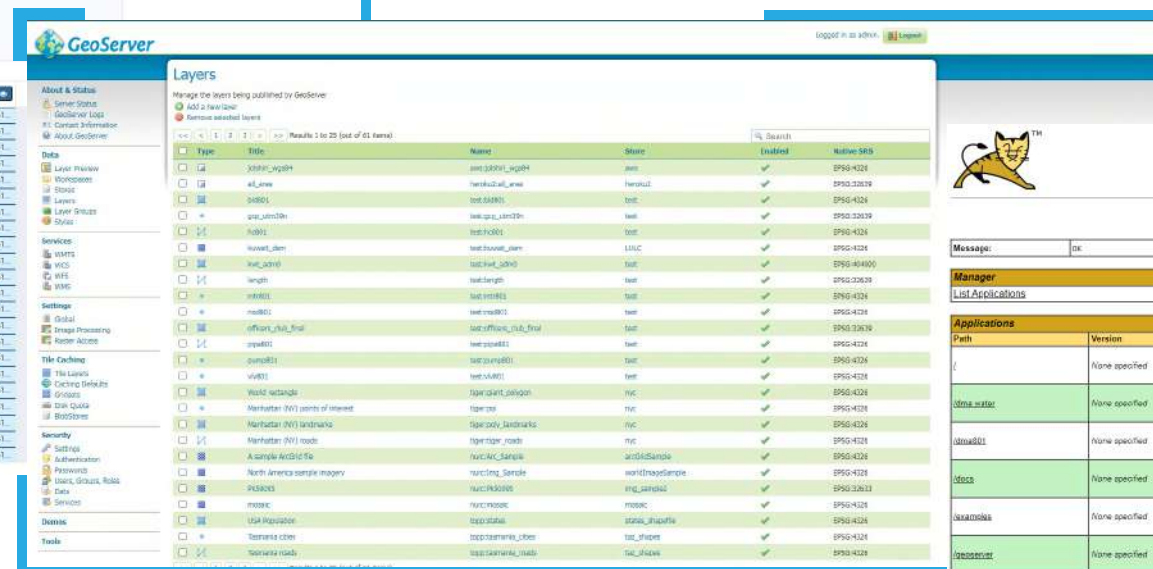
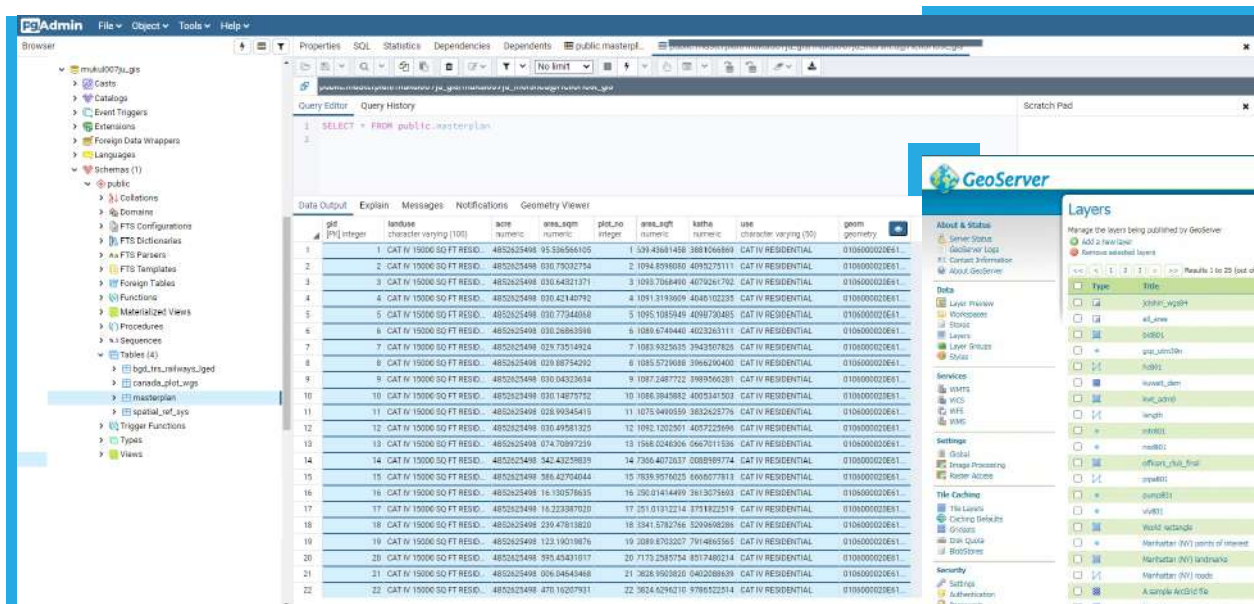
Postgre SQL database is a very powerfull database for carrying out spatial analysis and mapping. I have experience in creating database in remote server, data manipulation, spatial vector data and raster data processing, stroage and maintanance.



Geoserver is a web map server for WMS, WMTS, WFS. Geoserver is very stable web server for hosting gis layers. I have created few gis web application based on geoserver functionality. I have experience connecting postgis database to the server, layer publishing, styling, feature editing, WMS, WFS service publishing.

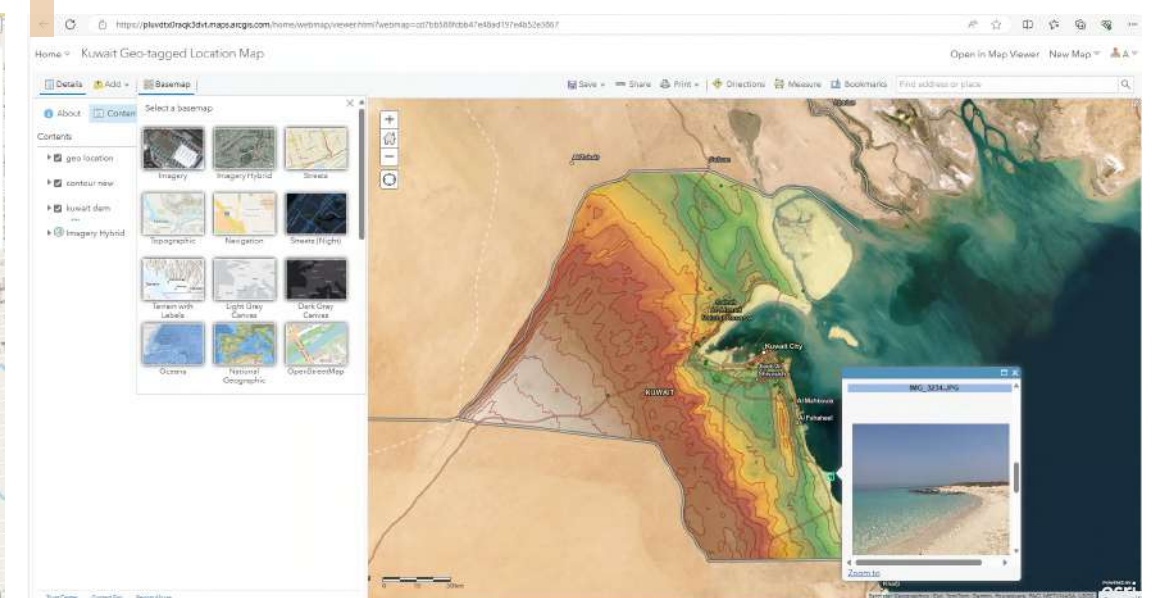
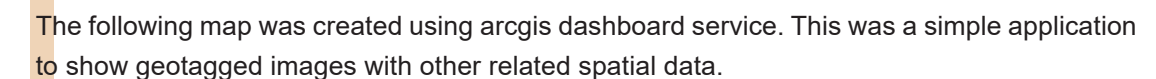
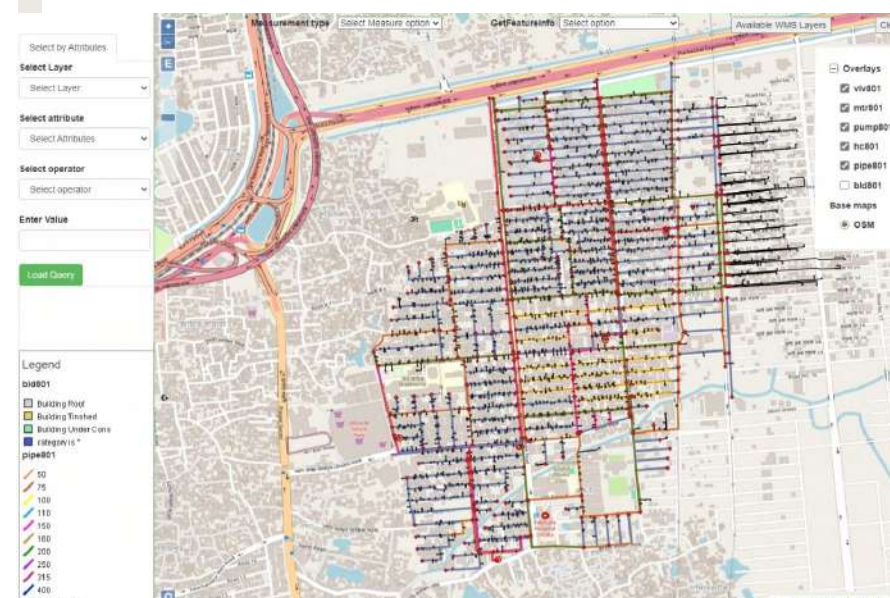
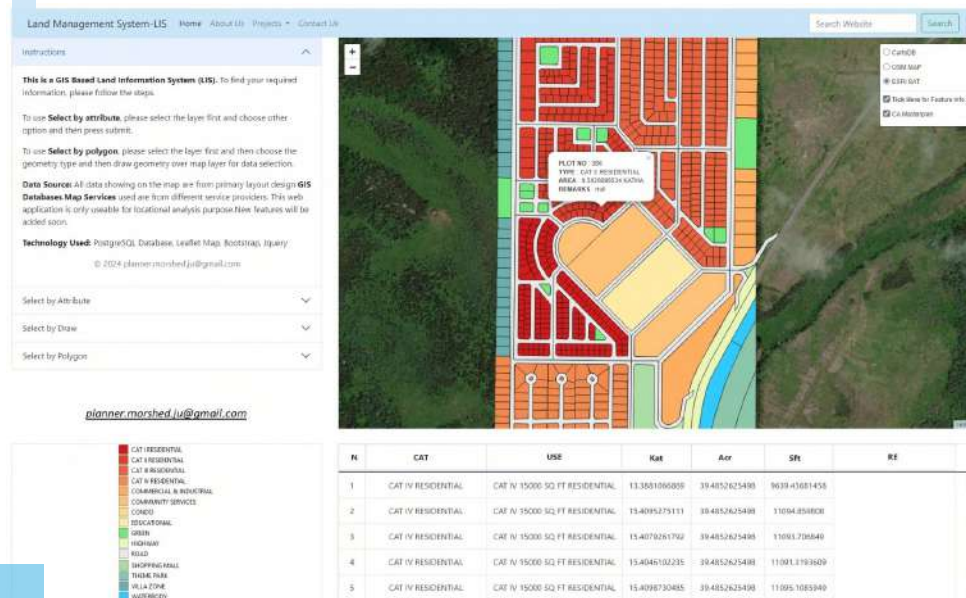


Tomcat server is for hosting java based web application. Best for web mapping application. Tomcat provides best support for hosting geoserver and it helps to create web map very easily.



I have prepared the following web map using postgis database. I have leaflet for web mapping and bootstrap for page design. This project is a land management system (LIS). You will find the site at: <https://mukul007ju.heliohost.us/>. This dashboard provides government information on land parcel and available plots for development. It will provide the real location of the proposed facilities in your neighbourhood.

The following web application was created using geoserver, postgis database and openlayers . The purpose of this app was for web utility mapping.



Web GIS Projects | Land Management Dashboard

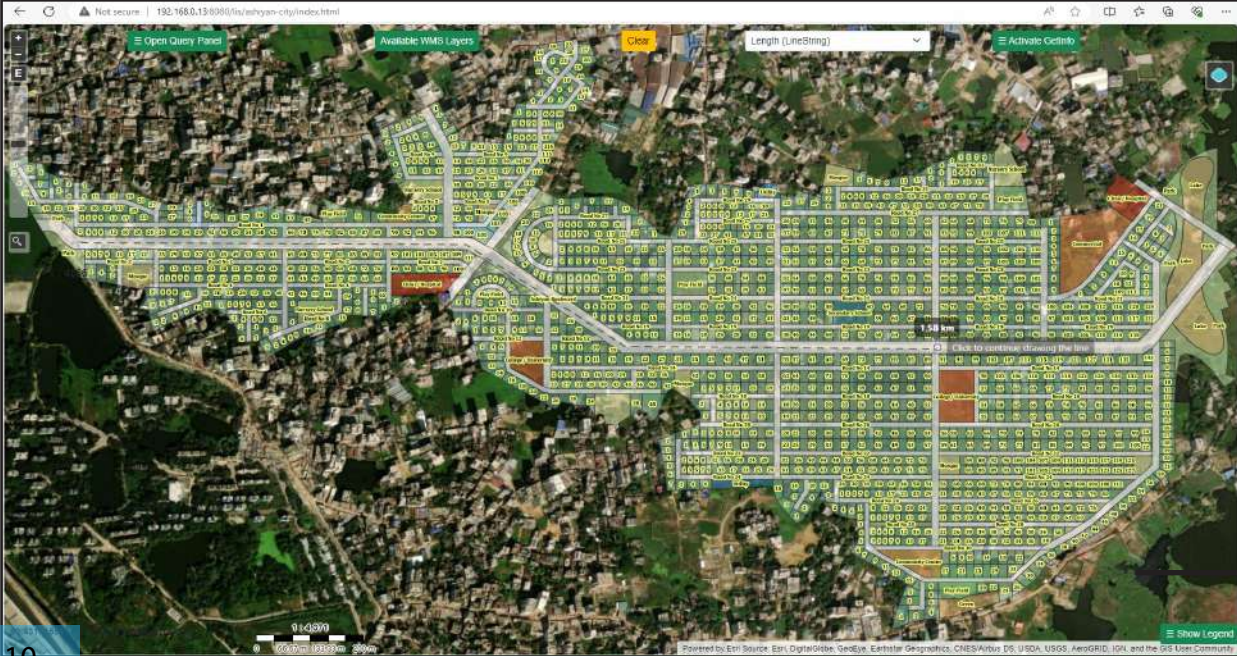
This project developed for a private residential land development company. It is designed to provide data sharing and map view to the users. There are many departments who requires information related to land purchase, government Mouza data, layout plan of the area, other related engineering information inside the project area.

For the implementation of the project; postgis database used for data storage, query and retrieving. Geoserver used for data styling, publishing and creating WMS service. And finally tomcat server provided the hosting service for the web application. Few javascript libraries have been used for mapping and designing of the website, like: OpenLayers and Bootstrap.

As for providing easy query support there is a query panel. If someone wants to know any information from a specific layer, just user need to select layer from the dropdown list and provide desired data, then it will be loaded in the table and highlighted in the map view. User will get information from four different layers; like: Proposed Layout Plan layer, DAP proposed Road and Waterbody layer, Government Mouza layer and Land Purchase Layer. There is a measurement panel for measuring any feature and find out area of any object. To facilitate interactive view, two different base maps have been provided. Users can switch from one basemap to another. An identify panel has been given to find whole details about a feature. A fixed table with scroll has been provided to represent features information together.

Following technologies have been provided the main support for this web GIS application.





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Load Layer

Query Panel

Measure Panel

Map View

Info Pop-Up

Legend

Table View

Select by AttributesSelect by Draw

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Load Layer

PostGIS data layers

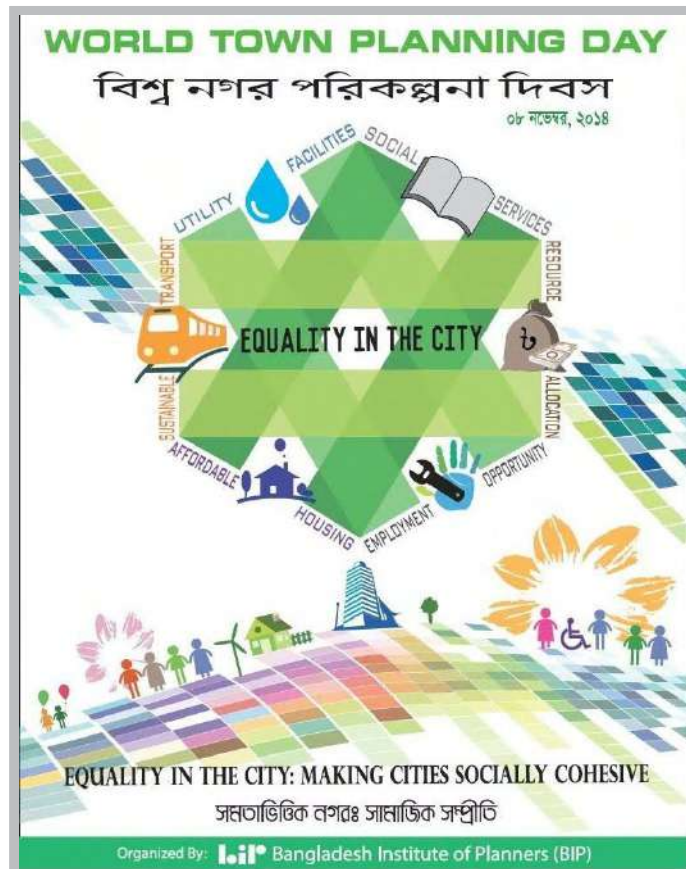
Open Street Base Map

ESRI Base Map

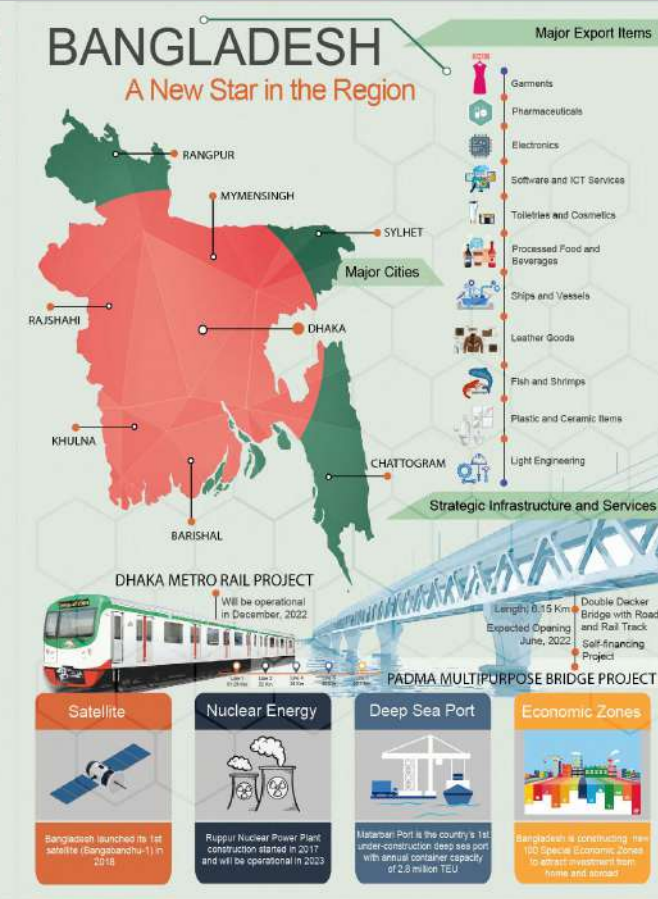
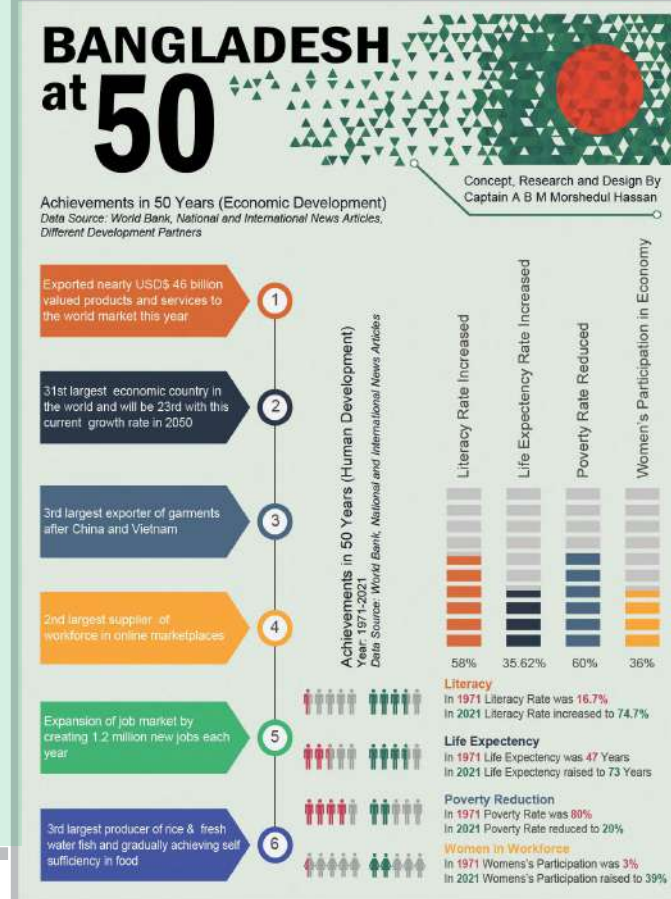
Map View

Table View

Design Projects | Posters, Infographic, Magazine Cover



This poster was designed for a competition for the celebration of World Town Planning Day 2014. I love to work with graphics to make people aware about city and their responsibilities for their city.



Bangladesh is a small country, but it has a lot to present. Bangladesh is doing good in international arena. This infographic was prepared to show the readers what Bangladesh has achieved during the last 50 years. This infographic was published in a issue of BMC Magazine, Kuwait.

BMC Magazine is a renowned publication by Bangladesh Military Contingent in Kuwait. The cover page of this issue was designed by myself to show the bonding between Bangladesh Armed Forces and Kuwait Armed Forces and celebrating independence of both countries.

I stood first for designing this poster for World Town Planning Day 2013. The concept was to show different type of decentralization through 3D graphics.

